

AI GUARDIAN · CYBERSECURITY × FINTECH × HEALTH

AEGIS

"Stop scams before they happen."

An AI guardian that hears manipulation in any conversation — voice, SMS, or chat — in real time, and intervenes *before* the money is lost: warning the user, looping in family, and freezing the transfer. One engine, every channel and language.

CYBERSECURITY	FINTECH	HEALTH
Social-engineering & deepfake intent	Real-time transaction shield	Protecting the vulnerable

SUBMISSION

Aegis — The AI Guardian That Stops Scams Before They Happen

LIVE DEMO

beyond-tomorrow-summit-ai-future-te.vercel.app

STATUS

Working prototype · live on free Groq Llama 3.3 70B · 61/61 tests pass

ENTRANT

Tumo Mogame — solo (full-stack, AI, design, pitch)

REPOSITORY

github.com/TUMO-MOGAME/Beyond-Tomorrow-Summit-AI-Future-Tech-Ecosystem-

CONTACT

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01 — THE PROBLEM

AI made scams infinitely scalable. Our defenses didn't keep up.

Generative AI made fraud faster, cheaper, and nearly impossible to detect — and the people who suffer most are those least equipped to fight back.

\$40B Projected US AI-enabled fraud losses by 2027 (up from \$12.3B in 2023 — 32% CAGR). DELOITTE	3 sec Of audio is enough to clone a voice at 85% accuracy. 70% of people can't tell it's fake. AARP
77% Of voice-clone scam victims lose money; 25% of adults have been affected or know someone who has. MCAFEE	5 min Average interval between deepfake fraud attempts — and accelerating. ENTRUST

WHY TODAY'S DEFENSES FAIL

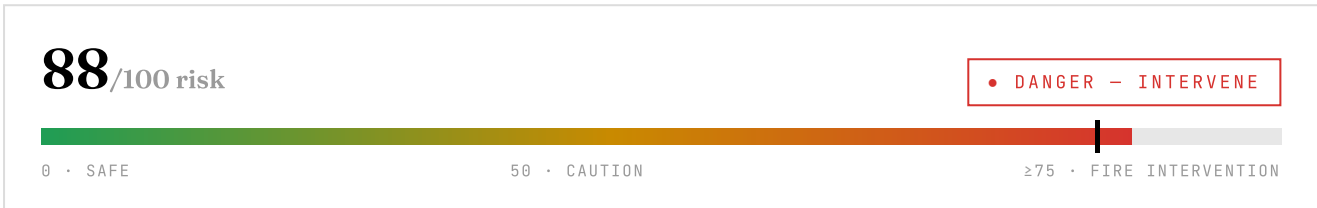
01	Reactive	Banks and existing tools flag fraud <i>after</i> the transfer clears. By then the money — often irreversible — is already gone.
02	Keyword-based	Blocklist and keyword filters are trivially evaded — AI-generated scams rewrite themselves on every attempt and walk straight past them.
03	No human moment	There is no intervention while the victim is being manipulated. They are alone with the attacker at the one moment that matters.

Everything today is forensics. The problem demands prevention.

02 — HOW IT WORKS

A real-time guardian that detects intent — and intervenes.

The Scam DNA Engine reads any conversation and detects the underlying manipulation structure — urgency, authority impersonation, secrecy, fear, irreversible payment — not surface keywords. When risk crosses the threshold on a money scam, Aegis acts.



- 1 Ingest — any channel**
A live call, SMS, or chat enters one modality-agnostic pipeline.
VOICE · SMS · CHAT
- 2 Scam DNA Engine**
A provider-agnostic LLM returns structured JSON: risk score, exact triggering quotes, manipulation tactics, a plain-language summary, and a recommended action.
GROQ · CLAUDE · ETC.
- 3 Warn + alert the Trusted Circle**
The user sees why it's dangerous in plain language; loved ones get a personalized email + SMS — the real-world circuit-breaker for elder fraud.
RESEND · TWILIO
- 4 Freeze the payment — Transaction Shield**
The suspicious transfer is held before it completes, stopping the loss at the moment it matters.
FINTECH
- 5 Guardian Number — real-world delivery**
No app can legally tap a live cellular call, so Aegis routes only *unknown* callers through a managed number, transcribes live, and on high risk *conferences in a loved one and holds the call*. Family calls are never monitored — privacy by design.
TWILIO VOICE

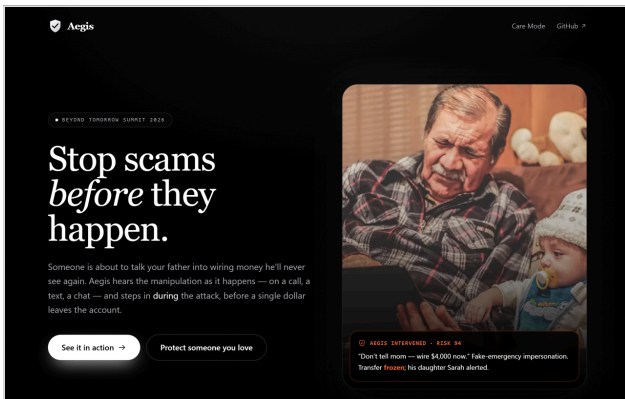
WHY AEGIS IS DIFFERENT — THE MOAT

01 Prevention, not forensics Acts <i>during</i> the attack, before the payment leaves — not after the money is gone.	02 Intent, not keywords One engine generalizes across every channel and language. The Spanish demo proves it — no retraining.	03 Trusted Circle + Care Mode Loops in family. The senior does nothing — family sets it up once. A social safety net, not a lonely alert.
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03 — WHAT THE JUDGES WILL SEE

Captured from the live product.

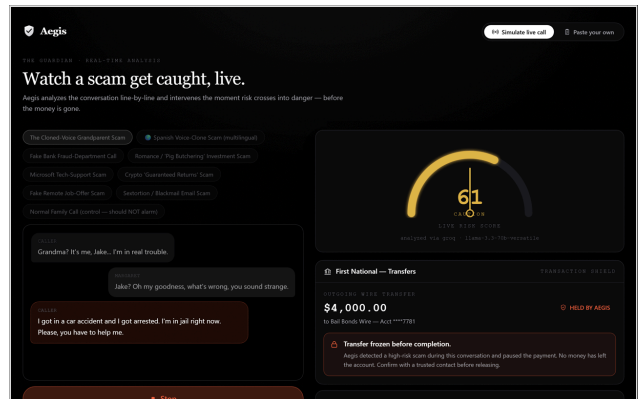
Real screens from the deployed app, running the real groq · llama-3.3-70b-versatile engine — not mockups. Monochrome on purpose: the people carry the warmth, the risk signal carries the only color.



01 · LANDING

The promise

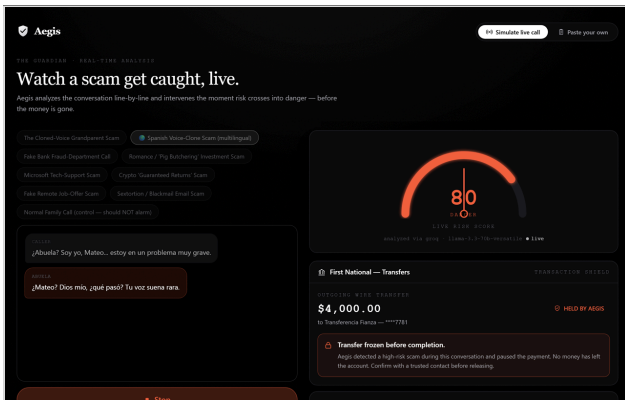
"Stop scams before they happen." A real person, the stakes stated plainly.



05 · GUARDIAN — PEAK

The catch

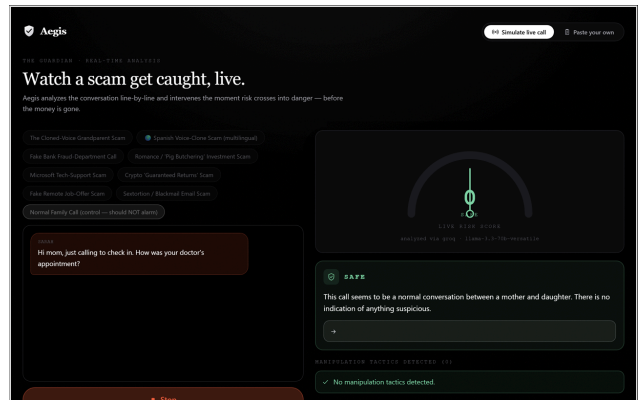
Risk peaks, tactics light up, the transfer is HELD, and the Trusted Circle is alerted.



06 · GUARDIAN — SPANISH

Intent, not keywords

The same engine flags a Spanish voice-clone scam — proof it reads intent across languages.



07 · GUARDIAN — SAFE

No false alarm

A normal family call scores SAFE — Aegis stays quiet when nothing is wrong.

04 — SEE IT IN 30 SECONDS

Run the rescue.

- 01 Open the live demo and go to the **Guardian** dashboard.
- 02 Hit **Simulate call** on the grandparent fixture — the scam transcript streams in.
- 03 Watch the **risk meter climb** and tactics light up with the exact words that triggered them.
- 04 At threshold, the **Trusted Circle** is alerted and the **Transaction Shield** freezes the transfer.
- 05 Run the **Spanish** fixture (language-agnostic) and the **safe** fixture (no false alarm).

BUILD STATUS

● 61 / 61 AUTOMATED CHECKS PASS

Prototype	End-to-end, live
Live engine	Groq Llama 3.3 70B (free)
Deploy	Vercel · public repo
Fail-safe	Offline mock — never hard-fails

PROVIDER-AGNOSTIC ENGINE

Auto-detects a key — Anthropic → Groq → Gemini → OpenRouter → OpenAI — with a tiered structured-output strategy and an offline heuristic **MOCK** fallback. Every integration degrades gracefully without a key.

THE CHALLENGE WE TURNED INTO THE ARCHITECTURE

We researched whether an app could simply "listen to grandma's calls" and hit a wall: **Apple forbids third-party access to live call audio, and Android is metadata-only** (plus a 2022 Play ban on accessibility-based recording). So we built the **Guardian Number** — route only unknown callers through a managed number, where live analysis, conferencing in family, and holding the call are all possible and legal with an up-front consent disclosure.

TECHNOLOGY STACK

AI / ML Provider-agnostic LLM — Groq Llama 3.3 70B (live) · Claude · Gemini · OpenRouter · OpenAI · structured output · Zod validation-with-retry
VOICE Twilio Programmable Voice — real-time transcription · conference · hold (Guardian Number) **ALERTS** Resend (email) · Twilio (SMS)
FRONTEND Next.js 14 · TypeScript · Tailwind · Framer Motion **BACKEND** Next.js API routes · Node.js **INFRA** Vercel · GitHub

WHAT'S NEXT

- GUARDIAN NUMBER** Productionize the Twilio prototype: screen real unknown callers; carrier conditional call-forwarding; low-latency intervention.
- PAYMENT HOLD** Bank / fintech partnerships (or Plaid-based monitoring) to freeze real transfers.
- PRIVACY** On-device / edge inference — conversations never leave the phone.
- REACH** B2B2C with banks, telcos & insurers; an embeddable SDK; eldercare caregiver dashboard.

"Scammers got an AI. It's time the rest of us got one too. We turn a \$40-billion problem into a moment of safety."

